

ECO 362 Financial Investments

Precept 2

September 15 – 18, 2025

- Bonds: No Arbitrage and Present Value Formula
- Stocks: No Arbitrage and Stochastic Discount Factor
- Cumulative Return

1 Bonds: No Arbitrage and Present Value Formula

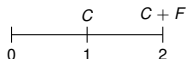
2 Stocks: No Arbitrage and Stochastic Discount Factor

3 Cumulative Return

A Simple Example

Consider the following bond market:

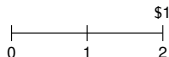
- A coupon bond with 2 years to maturity, annual coupon C and face value F , priced at P_c .



- A zero-coupon bond with 1 year to maturity (\$1 of face value), priced at $P_b(1)$.



- A zero-coupon bond with 2 years to maturity (\$1 of face value), priced at $P_b(2)$.



Replicating portfolio: C units of one-year zero plus $C + F$ units of two-year zero match the cash flows of the coupon bond. Then no arbitrage theory implies

$$P_c = C \cdot P_b(1) + (C + F) \cdot P_b(2) = \sum_{t=1}^T C \cdot P_b(t) + F \cdot P_b(T) \quad (1)$$

Or in terms of yields

$$P_c = \frac{C}{1 + Y_b(1)} + \frac{C + F}{[1 + Y_b(2)]^2} = \sum_{t=1}^T \frac{C}{(1 + Y_b(t))^t} + \frac{F}{(1 + Y_b(T))^T} \quad (2)$$

A More General Case

Consider the following asset market:

- An asset generating riskless cash flows X_t annually from year 1 to T (e.g., an annuity), priced at P .
- A set of zero-coupon bonds with 1 to T years of maturity (\$1 face value), priced at $P_b(t)$ for maturities $t = 1, 2, \dots, T$.

Replicating portfolio: X_t units of t -year zeros match the cash flows of the asset. Then no arbitrage theory implies

$$P = \sum_{t=1}^T X_t \cdot P_b(t). \quad (3)$$

Or in terms of annually compounded yields,

$$P = \sum_{t=1}^T \frac{X_t}{[1 + Y_b(t)]^t}. \quad (4)$$

Caution!

- The present value formula (derived from no arbitrage theory) **predicts** the relation between an asset (e.g., a coupon bond) and a set of zero-coupon bonds:

$$P_c = \sum_{t=1}^T C \cdot P_b(t) + F \cdot P_b(T) \quad (5)$$

$$P_c = \sum_{t=1}^T \frac{C}{[1 + Y_b(t)]^t} + \frac{F}{[1 + Y_b(T)]^T} \quad (6)$$

- The price-yield equation **defines** the relation between the price of a bond to its own yield:

$$P = \sum_{t=1}^T \frac{C}{(1 + Y)^t} + \frac{F}{(1 + Y)^T}, \quad (7)$$

where P is the price of a bond with T years to maturity, annual coupon C , face value F and annually compounded yield Y .

Par Yield

- It is the *coupon rate* at which a coupon bond would be priced at face value.
- Suppose the bond has face value \$1 and T years to maturity. Denote the par yield as $c(T)$.
- Suppose no arbitrage condition holds, then the present value formula implies

$$1 = \sum_{t=1}^T c(T) \cdot P_b(t) + 1 \cdot P_b(T)$$

- Solution:

$$c(T) = \frac{1 - P_b(T)}{\sum_{t=1}^T P_b(t)} \quad (8)$$

where $P_b(t)$ is the price of a zero-coupon bond with t years to maturity and \$1 of face value.

Forward Rate

- A forward contract locks in an interest rate (called a forward rate) on a future loan.
- Suppose you enter a forward contract and borrow \$1 in year T_1 with (annually compounded) forward rate $Y_f(T_1, T_2)$ until year T_2 :



- Now we invoke the no arbitrage argument, and note that the price of the forward contract is zero:

$$0 = \frac{1}{[1 + Y_b(T_1)]^{T_1}} + \frac{-[1 + Y_f(T_1, T_2)]^{(T_2-T_1)}}{[1 + Y_b(T_2)]^{T_2}} \quad (9)$$

which implies

$$Y_f(T_1, T_2) = \left\{ \frac{[1 + Y_b(T_2)]^{T_2}}{[1 + Y_b(T_1)]^{T_1}} \right\}^{\frac{1}{T_2-T_1}} - 1, \quad (10)$$

where $Y_b(T)$ is the (annually compounded) T -year zero-coupon yield.

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Replicating Portfolio

- Unlike (Treasury) bonds, stocks have risky cash flows: the payoffs at time t , $X_t(s)$, depends on the state of the world s .
- Pricing by no arbitrage: construct a replicating portfolio, matching the cash flows of the target asset in every state, then the price of the target asset must equal the price of the portfolio.
- One-period two-state example:

Stock	A	B	C
State 1	100	100	200
State 2	0	100	50
Price	20	50	?

- Strategy: w_A units of stock A + w_B units of stock B :

$$\begin{cases} 100w_A + 100w_B = 200 \\ 0w_A + 100w_B = 50 \end{cases} \implies \begin{cases} w_A = 1.5 \\ w_B = 0.5 \end{cases} \implies P_C = w_A P_A + w_B P_B = 55. \quad (11)$$

- If the market price of stock C were higher than \$55, buying the replicating portfolio and selling stock C would give you arbitrage profits!

State Prices

- Pricing by state prices: find the state prices from given asset payoffs and prices, then use these to evaluate the payoffs of the target asset.
- The same one-period two-state example:

Stock	A	B	C
State 1	100	100	200
State 2	0	100	50
Price	20	50	?

- Suppose price of \$1 in state 1 is m_1 , in state 2 is m_2 :

$$\begin{cases} 100m_1 + 0m_2 = 20 \\ 100m_1 + 100m_2 = 50 \end{cases} \implies \begin{cases} m_1 = 0.2 \\ m_2 = 0.3 \end{cases} \implies P_C = 200m_1 + 50m_2 = 55. \quad (12)$$

Stochastic Discount Factor

- Pricing by stochastic discount factor: find the SDF from given asset payoffs and prices and state probabilities, then use these to evaluate the payoffs of the target asset.
- Yet the same one-period two-state example:

Stock	A	B	C	Prob
State 1	100	100	200	0.6
State 2	0	100	50	0.4
Price	20	50	?	

- Suppose SDF in state 1 is M_1 , in state 2 is M_2 :

$$\begin{cases} 0.6 \times 100M_1 + 0.4 \times 0M_2 = 20 \\ 0.6 \times 100M_1 + 0.4 \times 100M_2 = 50 \end{cases} \implies \begin{cases} M_1 = 0.33 \\ M_2 = 0.75 \end{cases} \quad (13)$$

Hence, $P_C = 0.6 \times 200M_1 + 0.4 \times 50M_2 = 55$.

Relationship between State Prices and Stochastic Discount Factor

- Stochastic Discount Factor for state s

$$M(s) = m(s)/\phi(s)$$

where $\phi(s)$: probability of state s happening

- The price of asset i

$$\begin{aligned} P_i &= \sum_{s=1}^S m(s)X_i(s) \\ &= \sum_{s=1}^S \phi(s)M(s)X_i(s) \\ &= E[MX_i] \end{aligned}$$

SDF $M(s)$ is used as a **discounting factor** for the risky cash flow at state s , $X_i(s)$.

Stochastic Discount Factor

- A general pricing formula:

$$P = \sum_{t=1}^{\infty} \mathbb{E}[M_t X_t], \quad (14)$$

- Zero-coupon bond as a special case, $X_T = 1$ for T -year ZCB:

$$P_b(T) = \mathbb{E}[M_T]. \quad (15)$$

- Another way to look at coupon bonds through SDF,

$X_t = C, t = 1, 2, \dots, T - 1$ and $X_T = C + F$:

$$P_c = \sum_{t=1}^T \mathbb{E}[M_t \cdot C] + \mathbb{E}[M_T \cdot F] \quad (16)$$

$$= \sum_{t=1}^T \mathbb{E}[M_t] \cdot C + \mathbb{E}[M_T] \cdot F \quad (17)$$

$$= \sum_{t=1}^T P_b(t) \cdot C + P_b(T) \cdot F \quad (18)$$

Idiosyncratic vs Systematic Risks

- A decomposition of the pricing formula:

$$P = \sum_{t=1}^T \mathbb{E}[M_t X_t] \quad (19)$$

$$= \sum_{t=1}^T \left(\mathbb{E}[M_t] \mathbb{E}[X_t] + \text{Cov}(X_t, M_t) \right) \quad (20)$$

$$= \sum_{t=1}^T \left(P_b(t) \mathbb{E}[X_t] + \text{Cov}(X_t, M_t) \right) \quad (21)$$

- Two types of risk:
 - Idiosyncratic (diversifiable) risk: $\text{Cov}(X, M) = 0$
 - Systematic (non-diversifiable) risk: $\text{Cov}(X, M) \neq 0$
- If there is only idiosyncratic risk, price equals present value of expected cash flows.

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- An **asset i** pays a stream of random payoffs $X_{i,t}$, $t = 1, \dots, +\infty$
- Sell the asset at price $P_{i,t}$ at period t , after collecting the payoff $X_{i,t}$
- What is the **return of asset i** at period t ?
 - " the rate of return if you purchase the asset at period $t - 1$ and sell it at period t "

$$r_{i,t} = \frac{(P_{i,t} + X_{i,t})}{P_{i,t-1}} - 1$$

- Calculate cumulative return from monthly return:

$$1 + r_{cumulative,T} = \prod_{t=1}^T (1 + r_{i,t})$$

- For portfolio, calculate the portfolio monthly return first:

$$r_{p,t} = \sum_i w_{i,t-1} r_{i,t}.$$

Summation is over all assets in the portfolio.

- Example: two assets: risk-free bond and stock, with weight $1 - w_{t-1}$ and w_{t-1}

$$r_{p,t} = (1 - w_{t-1})r_{b,t} + w_{t-1}r_{s,t}.$$

- Let $r_{b,t}$ and $r_{s,t}$ be the returns of bond and stock respectively.
- A portfolio with proportion w of wealth in the stock has return:

$$r_{p,t} = (1 - w)r_{b,t} + w \cdot r_{s,t}$$

- Expected return:

$$\mathbb{E}[r_{p,t}] = r_{b,t} + w(\mathbb{E}[r_{s,t}] - r_{b,t})$$

- Return of bond is risk-free: no volatility in $r_{b,t}$. Portfolio volatility:

$$\sigma_{p,t} = w \cdot \sigma_{s,t}$$

- Capital market line (CML):

$$\mathbb{E}[r_{p,t}] = r_{b,t} + \frac{\mathbb{E}[r_{s,t}] - r_{b,t}}{\sigma_{s,t}} \sigma_{p,t}$$

- Comparison of cumulative returns of stocks and bonds
- Mean and standard deviation